Reflective Journal: NewsBot Intelligence System

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github.com/Haryapandhita/ITAI2373-NewsBot-Midterm

Solo project: "Team Collaboration Analysis" is covered as Project Planning & Work Process, and "Individual Contributions" as My Contributions & Design Decisions.

1. Project Planning & Work Process

The approach adopted followed as closely as possible to the sequence outlined in the starter notebook, following step by step through the modules from set-up and data loading through to implementation of the NewsBot class. Before new code was written, the starter sections were examined in detail to check they worked with the real BBC data. This no doubt prevented many problems occurring at the last minute. A SAMPLE_MODE switch was introduced for initially running on 200 articles, and only after the entire notebook was successfully run was the final run carried out on all 1,432 articles. Due to the unpredictable dropout of Colab's free tier, the spaCy output was cached and intermediate results retained, and this helped on many occasions. I used Claude as a tutor and pair programmer to help write and explain code, troubleshoot, and also explore options with their trade-offs. I ran every cell, made the design decisions and wrote the interpretations myself, which Claude then checked against the output. The sample-first approach worked well. However, I think it might be better to run a full set earlier; some of the conclusions based on samples were wrong.

2. Technical Integration Challenges

Initially, a problem was encountered with respect to the data set itself. The entire BBC text was lowercase, with commas and apostrophes removed, already in the Kaggle version of the dataset. This was caught during the text-format verification step and affected nearly all subsequent analyses. The NLP library spaCy has a heavy reliance on capitalisation to detect named entities and proper nouns (NNP). An experiment conducted with one of the texts showed that bill gates did not get picked up, but when written as Bill Gates, did. There was no solution possible within the dataset to re-capitalise the text, so a decision was made to keep a raw version for spaCy and VADER, although the missing capital letters still limit the detection of proper nouns and people. This was noted instead of falsely claiming an ability for named entity recognition.

The second issue came from the starter code. Multinomial Naive Bayes wasn't working because VADER scores could be negative. In addition, TF-IDF was fitted on the whole dataset before the train/test split, which meant test vocabulary was included in train. The first problem was solved with scaling the numeric data to [0, 1] and the second was solved with doing the TF-IDF and scaler within a scikit-learn Pipeline and fitting on train only.

The third and possibly most difficult issue was the adjustment of conclusions based on sample and full dataset analysis. In the sample of 200 articles, additional features seemed damaging, the Business category seemed to be the main problem, and parameter tuning seemed advantageous. However, in the full 1432 dataset, sentiment and length features only slightly improved on the default, the Tech class seemed to be the poorest, and the default settings were already the best. Adjusting conclusions that were based on a small, misleading sample was interesting.

3. Business Value Assessment

The primary end users who might find the system most realistic are journalists, news readers who want a faster overview, and media experts who track media coverage in terms of companies or politicians. The system offers automatic categorization with a 96.2% test accuracy and several other features such

as entity and location tracking and the general sentiment and emotion analysis of each article. These features are very useful to quickly sort through lots of news articles. The system accuracy varies and is not excellent: technology (the category with the poorest performance based on the F1 scores) achieves an F1 score of just 0.920. VADER in fact saturates around ±1 for about 70% of whole articles and there are domain-specific conventions and idioms that the system does not handle (e.g. "slam," "penalty," "party"). Furthermore, the system makes mistakes in NER (e.g. identifying "eu" as a PERSON). Therefore this system might not be advisable for autonomous final decision-making. Articles with low confidence levels and articles where the top two categories are very close should always be handed over to a human. However, overall in practical use, the system would probably be valuable as a first-stage monitoring and sorting process in which users treat the sentiment as a rough, potentially inaccurate signal rather than a fact.

4. My Contributions & Design Decisions

Decisions on several aspects of the design helped improve the results; some decisions were made by me, others followed a comparison of options offered by Claude. 50 exact duplicates and a single article from each of the 8 near-duplicate pairs (58 in total) were eliminated, because a story appearing in both the training and the test data would inflate accuracy; a slightly lower but more trustworthy figure was preferred. Then, the sentence-average VADER score was used as the default sentiment measure because the whole-article score saturated on long articles, making it impossible to tell articles apart. A model was selected based on cross-validation rather than test accuracy, so the 287 held-out articles were only used once, at the final stage. And within NewsBot, NER was performed only within the article body, following tests that the Title-Case headlines caused spaCy to treat non-existent objects as real entities. The choice to eliminate duplicates is the one I value most, because it is the key to the trustworthiness of the 96.2% figure.

5. Future Enhancements

With time, at least three improvements emerge: Firstly, original casing seems quite important, as a lot of issues related to NER and POS tags are probably related to the fact that text is lowercased, and not about the models. Secondly, coreference resolution is also very important: 38% of the approximately 17,000 "who did what to whom" triples have an agent using a pronoun which might lose some relationships' clarity. Thirdly, entity linking seems fundamental in being able to link "UK" and "Britain" or "Blair" and "Tony Blair". Transformer models like BERT and a Gradio dashboard could also be useful, but original casing may be the most fundamental. Again, this is primarily a data issue and even state-of-the-art models cannot fully recover information that was removed before they see the text.

6. Professional Development Impact

This was my first experience of building a complete NLP pipeline from scratch (raw data to a class capable of parsing a new article). I developed skills in cross validation, how to avoid data leakage, and identifying issues for example with a f-string not being compatible with python 3.12, an infinite loop in the tree-depth function, and modifications to the nrclex API. This experience also taught me to not take my results at face value. Interaction with an AI tutor meant that I had to understand what each line of code was doing as I am the one who has to explain and defend it. There is also an ethical aspect to consider in that the sentiment analysis scores associated with people and companies could be interpreted as judgements even when based on otherwise neutral terms e.g. penalty in a sports report. I hope to present the project on GitHub including a README, with honest discussion of the project's limitations. In the future, I want to explore transformer-based NLP and model fine-tuning.